



Barker
College

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Student Number

2022

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 1

Staff Involved:

- GDH • GPF • JZT • DZP*
- KJL • ESP • AXD

Friday 12th August 2022

100 copies

General

Instructions:

- Reading time - 10 minutes
- Working time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated
- Write your Student Number at the top of this page, and on all writing booklets

Total marks:

70

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Allow about 20 minutes for this section.

Section II - 60 marks (pages 6 - 9)

- Attempt Questions 11 - 14
- Allow about 1 hour and 40 minutes for this section.
- Answer each question in a SEPARATE writing booklet.

Section I

10 marks

Attempt Questions 1 – 10

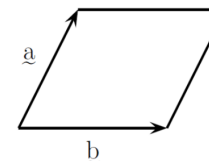
Allow about 20 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 The equation $x^3 + bx^2 + cx + d = 0$ has roots $1 + \sqrt{2}$, $1 - \sqrt{2}$ and 4. What is the value of b ?

- A. -6
- B. -4
- C. 4
- D. 6

- 2 The diagram below shows a rhombus, spanned by the two vectors $\underline{a}$ and $\underline{b}$.



Which of the following statements is true?

- A. $\underline{a} \cdot \underline{b} = 0$
- B. $(\underline{a} + \underline{b}) \cdot (\underline{a} - \underline{b}) = 0$
- C. $|\underline{a} + \underline{b}| = |\underline{a} - \underline{b}|$
- D. $2\underline{a} + 2\underline{b} = 0$

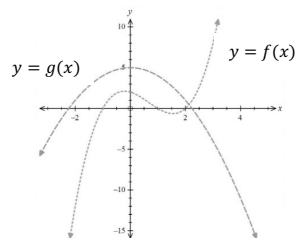
- 3 For what values of a and b does $y = e^{2x} + e^{-2x}$ satisfy the equation $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$?

- A. $a = -1, b = -2$
- B. $a = -2, b = 0$
- C. $a = 0, b = -4$
- D. $a = -2, b = -1$

- 4 Which is the vector resulting from the projection of $\vec{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ onto $\vec{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$?

- A. $\begin{pmatrix} \frac{-2}{\sqrt{5}} \\ \frac{-4}{\sqrt{5}} \end{pmatrix}$ B. $\begin{pmatrix} \frac{-2}{5} \\ \frac{-4}{5} \end{pmatrix}$
C. $\begin{pmatrix} \frac{8}{\sqrt{17}} \\ \frac{-2}{\sqrt{17}} \end{pmatrix}$ D. $\begin{pmatrix} \frac{8}{17} \\ \frac{-2}{17} \end{pmatrix}$

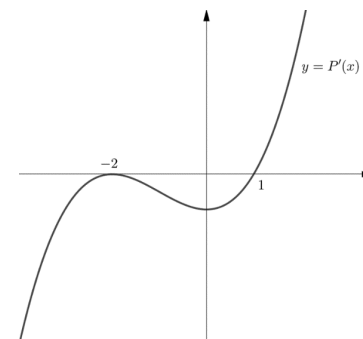
- 5 The graphs of $y = f(x)$ and $y = g(x)$ are shown.



Which diagram shows the graph of $y = f(x) + g(x)$?

- A. B. C. D.

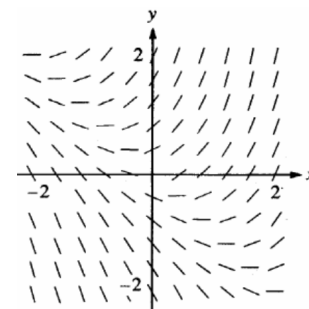
- 6 The graph of $y = P'(x)$ is shown, where $P(x)$ is a polynomial.



If $x = -2$ is a root of $P(x)$, what is the multiplicity of this root for $P(x)$?

- A. 0
B. 1
C. 2
D. 3

- 7 Which of the following differential equations best represents this direction field?



- A. $\frac{dy}{dx} = 1 + x$ B. $\frac{dy}{dx} = x^2$
C. $\frac{dy}{dx} = x + y$ D. $\frac{dy}{dx} = \frac{x}{y}$

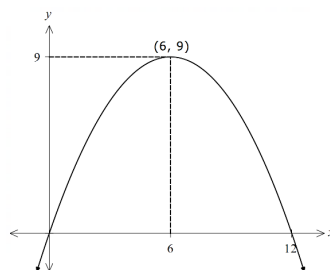
8 What is the value of $\sin^{-1}(\cos \theta)$, given $\frac{\pi}{2} < \theta < \pi$?

- A. $\frac{\pi}{2} - \theta$
- B. $\pi - \theta$
- C. $\frac{\pi}{2} + \theta$
- D. $\pi + \theta$

9 Ten cards numbered from 1 to 10 are laid face down. What is the least number of cards that must be picked to ensure that one of the numbers is twice the other?

- A. 6
- B. 7
- C. 8
- D. 9

10



Which of the parametric equations below represents the parabola above?

- A. $x = 12t, y = 9t^2$
- B. $x = 12t, y = 9 - t^2$
- C. $x = 6 - 2t, y = 9 - t^2$
- D. $x = 2t - 6, y = 9t - t^2$

End of Section I

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 40 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 - 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) [START A NEW BOOKLET]

(a) Solve the inequality $\frac{3x}{x-2} \geq 2$. 3

(b) Determine the coefficient of x^3 in the expansion of $(2-3x)^8$. 2

(c) Prove the identity $\sin(3x) + \sin x = 4 \sin x \cos^2 x$. 2

(d) A state senate sub-committee is to be formed by choosing 3 senators from among 6 rural senators and 6 urban senators. How many different sub-committees can be formed if at least 1 rural and 1 urban senator must be chosen? 2

(e) Find the exact value of $\int_0^3 \frac{1}{9+x^2} dx$. 1

(f) For the function $f(x) = \frac{\pi}{2} - \sin^{-1}\left(\frac{x}{3}\right)$

(i) State the domain and range of the function. 1

(ii) Sketch the function. 2

(g) Simplify $\frac{{}^{12}C_r}{{}^{11}C_{r-1}}$. 2

Question 12 (15 marks) [START A NEW BOOKLET]

- (a) Solve the initial value problem where $\frac{dy}{dx} = 2xe^{-y}$ given $y(0) = \ln 4$. 3
Express your answer with y as the subject.

- (b) The area bound by the curve $y = \sin^{-1} x$ and the y -axis between $y = 0$ and $y = \frac{\pi}{2}$ is rotated about the y -axis. Calculate, in exact form, the volume of the generated solid. 3

- (c) According to Newton's Law of Cooling, the rate at which a substance cools is proportional to the difference between the temperature of the substance and that of the surrounding air. This can be represented by the first order differential equation $\frac{dT}{dt} = k(T - A)$ where T is the temperature of the substance in degrees Celsius, t is the elapsed time in minutes, A is the air temperature and k is the cooling constant.

- (i) By solving the differential equation $\frac{dT}{dt} = k(T - A)$, show that $T = A + P_0 e^{kt}$ is a solution, where P_0 is a constant. 2
- (ii) If the temperature of the air is 20°C and a substance cools from 100°C to 50°C in 15 minutes, find the values of P_0 and k (to 4 decimal places). 2
- (iii) Hence find the time it will take for the temperature of the substance to cool from 100°C to 40°C . Round your answer to the nearest minute. 1

- (d) A projectile is launched across a level plain at 30° to the horizontal and at an initial speed of 60 m/s. Taking the origin as the launching point and $g = 10 \text{ m/s}^2$, the displacement vector $\underline{s}(t)$ for the projectile is given by:

$$\underline{s}(t) = (30\sqrt{3}t)\underline{i} + (30t - 5t^2)\underline{j} \quad (\text{Do NOT prove this})$$

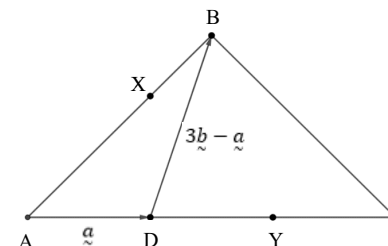
- (i) Find the velocity vector $\underline{v}(t)$ for the projectile. 1
- (ii) Find the maximum height of the projectile. 2
- (iii) Find the exact speed of the projectile one second after launch. 1

Question 13 (15 marks) [START A NEW BOOKLET]

- (a) By rewriting $\sin x + \cos x$ in the form $R \sin(x + \alpha)$, solve the equation: 3

$$\frac{\sin x + \cos x}{\sqrt{2}} = \frac{1}{2} \quad \text{for } 0 \leq x \leq 2\pi.$$

- (b) On the diagram below, $\overrightarrow{AD} = \underline{a}$, $\overrightarrow{DB} = 3\underline{b} - \underline{a}$ and AC and AB are straight lines. 3



Given that $AD : DY : YC = 1 : 1 : 1$ and $AX : XB = 2 : 1$, show that XY is parallel to BC .

- (c) Let the function $f(x) = \sec x$ be defined for the restricted domain $0 \leq x < \frac{\pi}{2}$, so that its inverse $f^{-1}(x)$ is also a function.

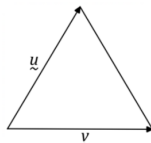
- (i) Sketch the graph of $f(x) = \sec x$ within $0 \leq x < \frac{\pi}{2}$. 1
- (ii) Show that $f^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$ and clearly state the domain of $f^{-1}(x)$. 2
- (iii) Find $\frac{d}{dx}(f^{-1}(x))$, writing your answer in **fully** simplified form. 2
- (iv) Hence or otherwise, describe the behaviour of the graph $y = f^{-1}(x)$ as $x \rightarrow \infty$. 1

- (d) A 10 metre ladder is leaning against a wall. The bottom of the ladder is sliding away from the wall at a rate of 4 m/s. Find the rate at which the top of the ladder is moving down the wall when the bottom of the ladder is 8 metres from the wall. 3

Question 14 (15 marks)

[START A NEW BOOKLET]

- (a) An equilateral triangle of side 4 units is shown below. Vectors u and v are represented in the diagram below. 1



Prove that $2|u - v| = u \cdot v$

- (b) Prove by Mathematical Induction that:

$$2(1!) + 5(2!) + 10(3!) + \dots + (n^2 + 1)n! = n(n+1)! \text{ for positive integers } n \geq 1 \quad 3$$

- (c) (i) Show that: $x^n(1+x)^n \left(1 + \frac{1}{x}\right)^n = (1+x)^{2n}$ 1
(ii) Hence prove that: 3

$$1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \binom{n}{3}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

- (d) (i) Find $\frac{d}{dx}(xe^x)$ 1
(ii) Using the substitution $u = e^x$ and the result from (i), find the exact value of: 3

$$\int_0^1 e^{2x} e^{(e^x)} dx$$

- (e) A total of five players are selected at random from four sporting teams. Each of the teams consists of ten players numbered from 1 to 10. 1
(i) What is the probability that of the five selected players, three are numbered '6' and two are numbered '8'? 1
(ii) What is the probability that the five selected players contain at least four players from the same team? 2

End of Paper

1. $2 \text{ roots} = -b$ $\therefore 1 + \sqrt{2} + 1 - \sqrt{2} + 4 = -b$ $6 = -b$ $b = -6 \therefore (A)$ 2. as a rhombus diagonals bisect at 90° $\therefore \text{Dot product} = 0$ $(a+b) \cdot (a-b) = 0 \quad (B)$ 3. $y = e^{2x} + e^{-2x}$ $y' = 2e^{2x} - 2e^{-2x}$ $= 2(e^{2x} - e^{-2x})$ $y'' = 4e^{2x} + 4e^{-2x}$ $= 4(e^{2x} + e^{-2x})$ Subbing into eq 1 $4e^{2x} + 4e^{-2x} + 2ae^{2x} - 2ae^{-2x} + be^x + be^{-2x} = 0$ $e^{2x}(4+2a+b) + e^{-2x}(4-2a+b) = 0$ $\therefore a=0, b=-4 \quad (C)$ 4. Prob $a = \frac{a \cdot b}{ b } b$ $= \frac{-2}{5} b$ $= \begin{pmatrix} -\frac{2}{5} \\ -\frac{4}{5} \end{pmatrix} = (B)$ 5. Add the y intercepts only possibility is (A) 6. Double root of $P(x)$ is a triple root of $f(x) \quad (D)$	7. Horizontal lines at line $x = -y \therefore \frac{dy}{dx} = x+y$ (C) $\therefore \text{Gradient } 0$ 8. $\sin^{-1}(\cos \theta) = \sin^{-1}(-\cos(\pi - \theta))$ not acute $= -\sin^{-1}(\cos(\pi - \theta))$ as odd $= -(\pi - \frac{\pi}{2} - (\pi - \theta))$ $= -(-\frac{\pi}{2} - \theta)$ $= \theta - \frac{\pi}{2} \quad (A)$ or sub in value of θ & check 9. Sets without doubles: $\left. \begin{array}{l} 134579 \\ 235789 \\ 2678910 \\ 1678910 \\ 135789 \end{array} \right\} \text{ all length 6}$ $\therefore \text{need 7 to ensure} \quad (B)$ 10. Check $(0,0)$ $A \ t = 0 \checkmark$ $B \ t = 0, 3 \times$ $C \ t = 3, 3 \checkmark$ $D \ t = 3, 0 \times$ Check $(12,0)$ $A \ t = 1, 0 \times$ $C \ t = -3, -3 \checkmark \therefore C$
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$$11) a) x \neq 2$$

$$3x(x-2) \geq 2(x-2)^2$$

$$3x^2 - 6x \geq 2x^2 - 8x + 8$$

$$x^2 + 2x - 8 \geq 0$$

$$(x+4)(x-2) \geq 0$$

$$x \leq -4, x \geq 2$$

b) General term

$$\binom{8}{k} 2^{8-k} (-3)^k$$

$$\therefore x^3 \text{ coeff} = \binom{8}{3} 2^5 (-3)^3$$

$$= -48384$$

$$c) LHS = \sin(x+2x) + \sin x$$

$$= \sin x \cos 2x + \cos x \sin 2x + \sin x$$

$$= \sin x (2\cos^2 x - 1) + 2\sin x \cos^2 x + \sin x$$

$$= 4\sin x \cos^2 x = RHS$$

$$d) {}_{12}P_4 = \binom{6}{2} \times \binom{6}{1}$$

$$= 90$$

$${}_{12}P_4 = 90 \therefore \text{Total } 180$$

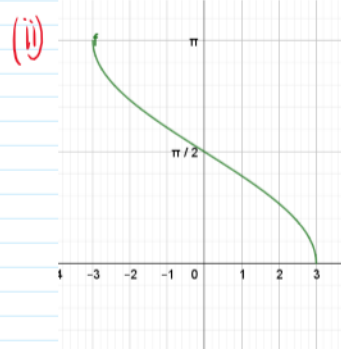
$$e) = \left. \frac{1}{3} \tan^{-1} \frac{x}{3} \right|_0^3$$

$$= \frac{\pi}{12} - 0$$

$$= \frac{\pi}{12}$$

$$f) (i) D: -3 \leq x \leq 3$$

$$R: 0 \leq y \leq \pi$$



$$g) \frac{{}_{12}C_r}{{}_{11}C_{r-1}} = \frac{12!}{r!(12-r)!} \div \frac{11!}{(r-1)!(12-r)!}$$

$$= \frac{12!}{r!(12-r)!} \times \frac{(r-1)!(12-r)!}{11!}$$

$$= \frac{12}{r}$$

$$12. a) dy e^y = 2x dx$$

$$\int e^y dy = \int 2x dx$$

$$e^y = x^2 + C$$

$$\text{Sub in } 0, \ln 4$$

$$\therefore 4 = 0 + C$$

$$C = 4$$

$$e^y = x^2 + 4$$

$$y = \ln(x^2 + 4)$$

b)

$$V = \pi \int_0^{\pi} x^2 dy$$

$$y = \sin^{-1} x$$

$$x = \sin y$$

$$x^2 = \sin^2 y$$

$$\therefore V = \pi \int_0^{\pi} \sin^2 y dy$$

$$= \frac{\pi}{2} \int_0^{\pi} (1 - \cos 2y) dy$$

$$= \frac{\pi}{2} \left[y - \frac{\sin 2y}{2} \right]_0^{\pi}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{2} - 0 - (0 - 0) \right) \right]$$

$$= \frac{\pi^2}{4} u^2$$

$$d) (i) \underline{v}(t) = 30\sqrt{3} \underline{i} + (30 - 10t) \underline{j}$$

$$(ii) \text{ max height when } j = 0$$

$$\therefore 30 - 10t = 0$$

$$t = 3$$

$$\therefore \text{ height} = 30 \times 3 - 5 \times 3^2$$

$$= 45 \text{ m}$$

$$c) (i) \frac{dT}{dt} = k(T-A)$$

$$\frac{dT}{k(T-A)} = dt$$

$$\int \frac{dT}{k(T-A)} = \int dt$$

$$\frac{1}{k} \ln|T-A| = t + C$$

$$\ln|T-A| = k(t+C)$$

$$T-A = e^{kt+kC}$$

$$T = A + e^{kt+kC}$$

$$= A + e^{kt} \times e^{kC}$$

$$e^{kC} \text{ is a constant}$$

$$\therefore T = A + P_0 e^{kt}$$

$$(ii) A = 20$$

$$t = 0, T = 100$$

$$\therefore P_0 = 80$$

$$t = 15, T = 50$$

$$\therefore 30 = 80 e^{15k}$$

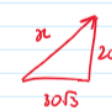
$$k = \frac{\ln \frac{3}{8}}{15} = -0.0654$$

$$(iii) T = 40 = 20 + 80 e^{kt}$$

$$t = 21.2 \text{ min}$$

$$= 21 \text{ min}$$

$$(iv) t = 1, \underline{v} = 30\sqrt{3} \underline{i} + 20 \underline{j}$$



$$x = \sqrt{3100} \text{ m/sec}$$

$$= 10\sqrt{31} \text{ m/sec}$$

$$13 a) \sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4})$$

$$\frac{\sin x + \cos x}{\sqrt{2}} = \frac{1}{2}$$

$$\sin x + \cos x = \frac{1}{\sqrt{2}}$$

$$\therefore \sqrt{2} \sin(x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}} \quad 45^\circ \leq x \leq 135^\circ$$

$$\sin(x + \frac{\pi}{4}) = \frac{1}{2}$$

$$x + \frac{\pi}{4} = \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$x = \frac{7\pi}{12}, \frac{23\pi}{12}$$

$$b) \vec{AB} = \vec{a} + 3\vec{b} - \vec{a}$$

$$= 3\vec{b}$$

$$\therefore \vec{AX} = 2\vec{b}$$

$$\vec{AB} = \vec{a} = \vec{OY} = \vec{YC}$$

$$\vec{XY} = -2\vec{b} + 2\vec{a}$$

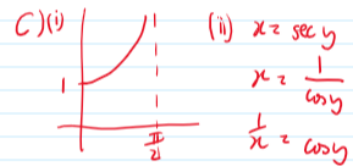
$$= 2(\vec{a} - \vec{b})$$

$$\vec{BC} = \vec{a} - 3\vec{b} + 2\vec{a}$$

$$= 3(\vec{a} - \vec{b})$$

$$\therefore \vec{XY} \text{ and } \vec{BC} \text{ are parallel}$$

as they are scalar multiples of each other

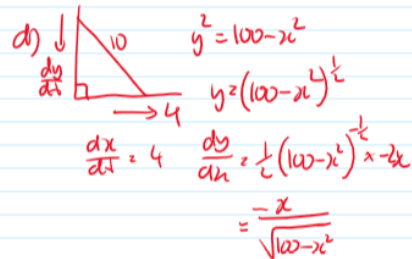


$$0 \leq x \leq 1$$

$$(iii) \frac{d}{dx} \cos^{-1} \frac{1}{x} = - \frac{-x^{-2}}{\sqrt{1 - (\frac{1}{x})^2}}$$

$$= \frac{1}{x\sqrt{x^2 - 1}}$$

(iv) from (iii) no turning points
 y approaches $\frac{\pi}{2}$ as
 $x \rightarrow \infty$



$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$= \frac{-x}{\sqrt{100 - x^2}} \times 4$$

$$= \frac{-4x}{\sqrt{100 - x^2}}$$

$$\text{when } x = 8$$

$$\frac{dy}{dt} = \frac{-32}{6}$$

$$= -\frac{16}{3} \text{ m/s}$$

$$\therefore \frac{16}{3} \text{ m/s down the wall}$$

$$14 a) \text{ RHS} = \underline{u \cdot v}$$

$$u \cdot v = |u||v| \cos \theta$$

$$= 16 \cos 60^\circ$$

$$= 8 \text{ units}$$

$$\text{LHS} = 2(u \cdot v)$$

$$= 2 \times 4$$

$$= 8$$

$$b) \text{ Let } n=1 \quad \text{LHS} = 2$$

$$\text{RHS} = 2$$

Assume $n=k$

$$2(1!) + 2(2!) + 10(3!) + \dots + (k+1)k! = k(k+1)!$$

Let $n=k+1$

RTP

$$2(1!) + 2(2!) + 10(3!) + \dots + (k+1)k! + (k+1)(k+1)! = (k+1)(k+2)!$$

$$\text{LHS} = k(k+1)! + (k+1)(k+1)(k+1)!$$

$$= (k+1)! (k+1)(k+2)$$

$$= (k+1)! (k+2)(k+1)$$

$$= (k+1)(k+2)! = \text{RHS}$$

$\therefore$ True by PMI

$$c) (i) \text{ LHS} = x^n (1+x)^n \left(\frac{x+1}{x}\right)^n$$

$$= (1+x)^n (1+x)^n$$

$$= (1+x)^{2n}$$

$$(ii) (1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

$$\left(1 + \frac{1}{x}\right)^n = 1 + \binom{n}{1}\frac{1}{x} + \binom{n}{2}\frac{1}{x^2} + \dots + \binom{n}{n}\frac{1}{x^n}$$

$$(1+x)^{2n} = 1 + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + \binom{2n}{n}x^n + \dots + \binom{2n}{2n}x^{2n}$$

Looking at coefficients of x^n

$$\text{LHS} = 1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \binom{n}{3}^2 + \dots + \binom{n}{n}^2$$

$$\text{RHS} = \binom{2n}{n}$$

$$\therefore 1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \binom{n}{3}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

Qn 14 ii) $LHS = x^n (1+x)^n \left(1 + \frac{1}{x}\right)^n$
 $= x^n \left({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n \right) \left({}^nC_0 + {}^nC_1 \frac{1}{x} + {}^nC_2 \frac{1}{x^2} + \dots + {}^nC_n \frac{1}{x^n} \right)$

To get coefficients of x^n , $(1+x)^n \times \left(1 + \frac{1}{x}\right)^n$ must be a constant.

That happens when you multiply their 1st terms together, 2nd terms, ..., last terms.

e.g. ${}^nC_0 \times {}^nC_0 = \binom{n}{0}^2$

${}^nC_1 x \times {}^nC_{n-1} \frac{1}{x} = \binom{n}{1}^2$

$\vdots$

$\binom{n}{n} x^n \times \binom{n}{n} \frac{1}{x^n} = \binom{n}{n}^2$

RHS: Coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$

$\binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + \binom{2n}{n}x^n + \dots + \binom{2n}{2n}x^{2n}$

Equating coefficients of x^n :

$\therefore 1 + \binom{2n}{1}^2 + \binom{2n}{2}^2 + \binom{2n}{3}^2 + \dots + \binom{2n}{n}^2 = \binom{2n}{n}$

d) (i) $u = x, v = e^x$
 $u' = 1, v' = e^x$
 $\therefore \frac{d}{dx} x e^x = x e^x + e^x$
 $= e^x (x+1)$

e) (i) $P = \frac{{}^4C_3 \times {}^4C_2}{{}^{40}C_5}$
 $= \frac{1}{27417}$

(ii) $u = e^x$
 $\frac{du}{dx} = e^x$
 $x=1, x=0$
 $u=e, u=1$
 $\int_1^e u^2 e^u \frac{du}{u}$ (ii) $P = \frac{{}^{10}C_5 \times 4 + {}^{10}C_4 \times {}^{30}C_1 \times 4}{{}^{40}C_5}$

$= \int_1^e u e^u du$
 now $\int u e^u du + \int e^u = u e^u$ from (i) $= \frac{28}{703}$

$\int_1^e u e^u du = [u e^u - e^u]_1^e$
 $= e \times e^e - e^e - (e - e)$

$= e^{e+1} - e^e$

$= e^e (e-1)$